# Sayan Ghosh

+91 97590 91771 sayanghoshofc@gmail.com linkedin.com github.com leetcode.com

Cybersecurity-focused CS undergraduate building secure backend systems and APIs, applying OWASP-aligned secure coding and application security testing, and prototyping GenAI/RAG tooling for security workflows.

## Education

**VIT Bhopal University** **Bhopal, MP, India**
*B.Tech, Computer Science and Engineering (Cybersecurity)* *Aug. 2023 – May 2027*

**CBSE Board (Class XII)**
*Higher Secondary Certificate, Physics/Chemistry/Math/Computer Science* *April 2023*

## Projects

**SentinelX IDS – AI-Powered Intrusion Detection System** | *Python, FastAPI, React, SQLAlchemy, WebSockets*

- Architected an intrusion detection system with a FastAPI backend (60+ REST APIs), React dashboard, and real-time WebSocket alerting, secured with JWT auth, role-based access control, and an audit trail
- Designed a Sigma-style detection engine with 8 YAML rules mapped to MITRE ATT&CK, a sliding-window event correlator, and statistical anomaly detection (Z-score, IQR) across 5 log parsers
- Integrated 5 threat-intel feeds with GeoIP enrichment and risk scoring, a multi-provider LLM copilot for alert triage, and Slack/Discord/email notifications, with Continuous Integration (GitHub Actions) running automated unit tests

**CipherMind – Security-Focused RAG Platform** | *Python, FastAPI, ChromaDB, Groq (Llama 3.1), Hugging Face*

- Deployed a retrieval-augmented generation (RAG) platform on Render using FastAPI, ChromaDB vector storage, and Hugging Face embeddings to ground Llama 3.1 responses in retrieved document chunks
- Conducted application security testing and vulnerability assessment across 2 domain-specific knowledge bases (general and cybersecurity: OWASP Top 10, Burp log templates), evaluating a security-auditing persona against SQL injection, privilege escalation, and path traversal
- Added resilience with retry and backoff for embedding cold starts, API-key failover on rate limits, capped session memory, and environment-based secrets management

**Home Lab Threat Intelligence & Detection Dashboard** | *Python, Go, PostgreSQL, Docker, Scapy, scikit-learn*

- Built a real-time network monitoring system across 3 decoupled microservices: Python/Scapy packet capture, a Go REST API with input validation and parameterized SQL, and PostgreSQL with INET types
- Implemented an unsupervised Isolation Forest pipeline over a 60-minute rolling window, using per-destination features (packet count, average size, total bytes) to flag anomalous endpoints such as potential C2 beaconing
- Delivered a SOC-style Streamlit dashboard with a 5-second refresh loop, containerized with Docker Compose

## Technical Skills

**Secure Software Engineering**: Python, Go, Java, JavaScript, C++, REST APIs, FastAPI, Unit Testing, CI/CD (GitHub Actions), Git
**AI & GenAI**: LLM APIs, Retrieval-Augmented Generation (RAG), Vector Databases (ChromaDB), Prompt Engineering, Hugging Face, Agent/Tool-Calling Workflows
**Application Security**: OWASP Top 10, Secure Coding, Threat Modeling, Vulnerability Assessment, JWT Auth, RBAC, MITRE ATT&CK
**Data & ML**: scikit-learn, pandas, NumPy, Isolation Forest, Statistical Anomaly Detection
**Cloud & Infrastructure**: Docker, AWS, Linux, PowerShell, PostgreSQL, MySQL, Redis

## Internships

**Software Developer Intern** **June 2026 – August 2026**
*UnsaidTalks Education Pvt. Ltd.* *Remote*

- Engineered the UnsaidTalks School of Technology (SOT) portal (sot.unsaidtalks.com) end-to-end, building modular NestJS backend services and Prisma ORM schemas that cut environment setup time by roughly 40%
- Shipped an AI-powered resume analyzer integrated into the student portal, delivering automated feedback to 300+ student resumes and improving profile completion rates by roughly 25%
- Launched a Campus Ambassador management portal using React, TanStack Query, and Router, connecting 50+ ambassadors across campuses within an Agile team running daily stand-ups

## Certifications & Achievements

Supervised Machine Learning – Coursera / DeepLearning.AI
Google IT Support Professional Certificate – Google / Coursera
Oracle Cloud Infrastructure Certified Data Science Professional – Oracle
Solved 300+ data structures and algorithms problems across LeetCode and other coding platforms